



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01207010**

MAY/JUNE 2015

FORM TP 2015044

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

BIOLOGY

Paper 01 - General Proficiency

1 hour 15 minutes

10 JUNE 2015 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
- (B) Diabetes
- (C) Influenza
- (D) Anaemia

Sample Answer

(A) (B) (C) ☒

The correct answer to this item is “Anaemia”, so (D) has been shaded.

5. If you want to change your answer, erase it completely and fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
7. Figures are not necessarily drawn to scale.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.



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Item 1 refers to the following data recorded by students who observed some organisms in their natural habitat.

	Organism			
	I	II	III	IV
Number of Body Segments	1	3	3	12
Number of Legs	4	8	6	24
Presence of Wings	1 pair	None	1 pair	None
Colour	Red	Blue	Black	Black

1. Which organism is MOST likely an insect?

- (A) I
- (B) II
- (C) III
- (D) IV

2. Which of the following organisms occupies the fourth trophic level in the food chain below?

Cabbage → Caterpillar → Small bird → Hawk

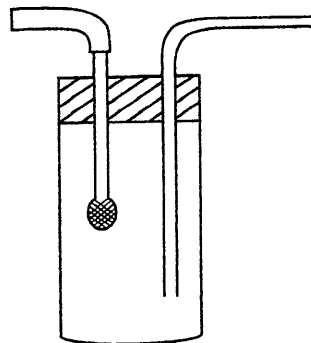
- (A) Hawk
- (B) Cabbage
- (C) Small bird
- (D) Caterpillar

3. Which of the following relationships may present harm to one of the organisms involved?

- I. Commensalism
- II. Parasitism
- III. Predation
- IV. Mutualism

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) III and IV only

Item 4 refers to the following diagram which shows an apparatus used for collecting organisms.



4. The apparatus is MOST suitable for collecting

- (A) ants
- (B) slugs
- (C) moths
- (D) spiders

5. Which of the following organisms are MOST important in biodegradation?

- (A) Bacteria
- (B) Vultures
- (C) Omnivores
- (D) Earthworms

6. Which of the following gases does NOT contribute to the greenhouse effect?

- (A) Methane
- (B) Carbon dioxide
- (C) Carbon monoxide
- (D) Chlorofluorocarbon

7. Replanting trees on a bare hillside is an example of

- (A) conservation and afforestation
- (B) preservation and conservation
- (C) restoration and preservation
- (D) reafforestation and restoration

8. Which of the following factors contribute to INCREASED population size?

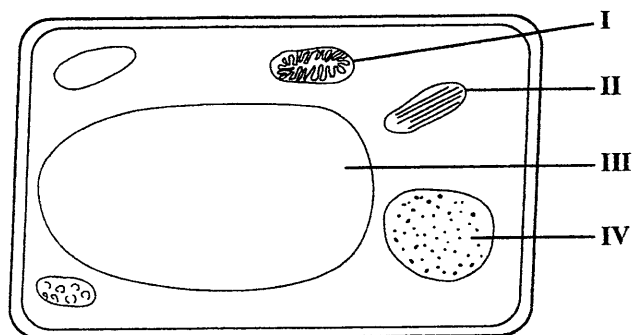
- I. Ability to resist disease
- II. Inadequate water supplies
- III. Too much or too little light
- IV. Small number of predators

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) I, II, III and IV

9. When compared to a cheek cell, a muscle cell contains more

- (A) vacuoles
- (B) ribosomes
- (C) mitochondria
- (D) chromosomes

Items 10–11 refer to the following diagram of a plant cell.



10. Which of the labelled structures is responsible for controlling cellular activities?

- (A) I
- (B) II
- (C) III
- (D) IV

11. Which of the labelled structures is NOT found in an animal cell?

- (A) I
- (B) II
- (C) III
- (D) IV

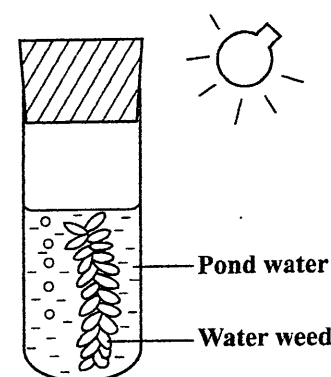
12. By which two processes do the dissolved minerals from the soil move into the roots of a plant?

- (A) Osmosis and diffusion
- (B) Active transport and osmosis
- (C) Diffusion and active transport
- (D) Transpiration pull and osmosis

13. Autotrophic nutrition does NOT occur in

- (A) moss
- (B) zooplankton
- (C) tomato plants
- (D) blue-green algae

Item 14 refers to the following diagram.



14. If the bubbles given off by the plants were collected and tested, the gas present would MOST likely be

- (A) oxygen
- (B) hydrogen
- (C) nitrogen
- (D) carbon dioxide

15. The intense yellowing of leaves is MOST likely due to a deficiency of

- (A) iron
- (B) calcium
- (C) magnesium
- (D) phosphorus

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Item 16 refers to the following table which shows the results of food tests performed on three food samples, X, Y and Z.

Food Sample	Test and Observation
X	Violet colour with sodium hydroxide and copper sulphate
Y	Blue/black precipitate with iodine
Z	Orange precipitate when heated with Benedict's reagent after hydrolysis

16. The samples X, Y and Z respectively MOST likely contain

	X	Y	Z
(A)	non-reducing sugar	starch	protein
(B)	protein	reducing sugar	starch
(C)	starch	protein	reducing sugar
(D)	protein	starch	non-reducing sugar

17. Excess amino acids in the diet are
- (A) passed out in the faeces
 - (B) converted to fat and stored
 - (C) converted to urea in the liver, then excreted
 - (D) taken directly to the bladder for excretion in urine

19. The surface area of the lungs is greatly increased by the

- (A) double membranes of the pleura
- (B) large numbers of alveoli
- (C) dense network of capillaries
- (D) branching of the bronchioles

Item 18 refers to the following information.

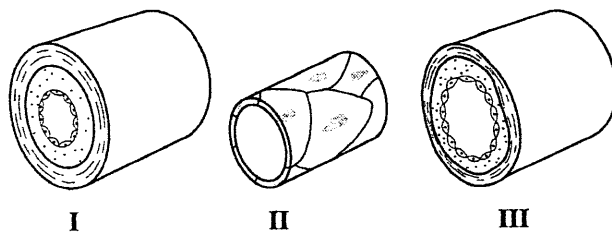
A man has poor night vision and his gums bleed whenever he brushes his teeth.

18. Which two vitamins are MOST likely lacking in his diet?
- (A) A and B₁
 - (B) B₁ and B₂
 - (C) C and A
 - (D) C and B₂

20. Which of the following is an effect of nicotine found in cigarette smoke?

- (A) Increased heart rate
- (B) Increased beating of cilia
- (C) Reduced oxygen transport by the blood
- (D) Increased mucus production in the cells lining the respiratory passage

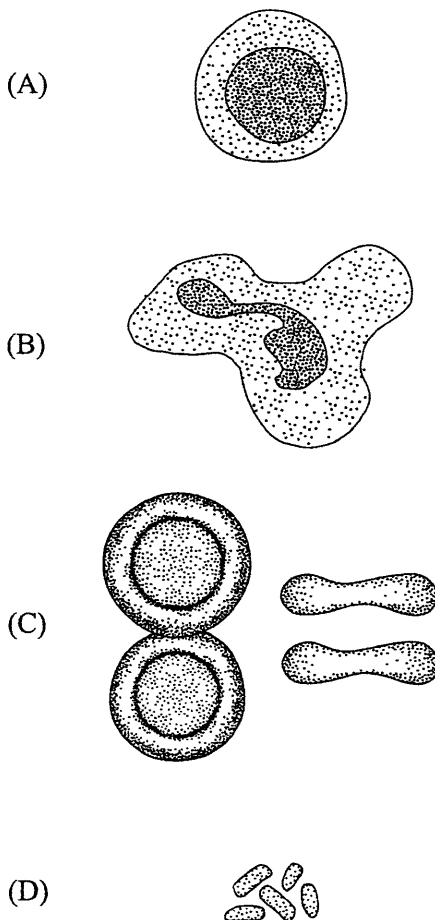
Item 21 refers to the following diagrams of blood vessels.



21. Which of the following correctly identifies the blood vessels?

	I	II	III
(A)	Vein	Capillary	Artery
(B)	Artery	Vein	Capillary
(C)	Capillary	Artery	Vein
(D)	Artery	Capillary	Vein

22. Which of the following components of blood could surround and engulf pathogens?



23. Which of the following are involved in the formation of blood clots?

- (A) Platelets
- (B) Antibodies
- (C) Phagocytes
- (D) Lymphocytes

24. Immunization is MOST effective in controlling the spread of a communicable disease if

- (A) a vaccine is administered to an uninfected individual
- (B) serum is administered to an uninfected individual
- (C) a vaccine is administered to an infected individual
- (D) serum is administered to an infected individual

25. Which of the following statements is NOT true of the xylem of a flowering plant?

- (A) It is non-living.
- (B) It contains mitochondria.
- (C) It has lignified walls for support.
- (D) It transports water and mineral salts.

26. Which of the following are reasons why transpiration is important to plants?

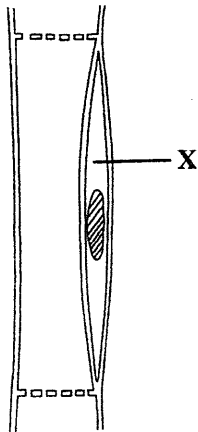
- I. Water and dissolved minerals are transported to the leaves.
- II. The leaves are cooled.
- III. Unwanted water is excreted from the leaves.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

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Item 27 refers to the following diagram which shows cells from the stem of a plant.



27. The part labelled X is the

- (A) cork cell
- (B) cambial cell
- (C) cortical cell
- (D) companion cell

28. Food is stored in the bulb of an onion plant to

- (A) provide food for humans
- (B) provide for the growth of new plants from the buds
- (C) prevent the plant from having to photosynthesize continuously
- (D) provide food for developing embryos in the next growing season

29. Which of the following would NOT represent a form of excretion in plants?

- (A) Gases diffusing through stomata
- (B) Tannins being stored in dead tissue
- (C) Water vapour diffusing through epidermis
- (D) Stored calcium oxalate crystals in leaves during leaf fall

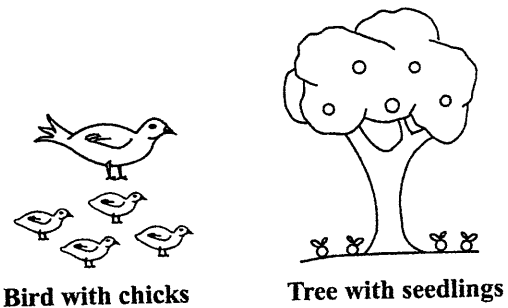
30. Although glucose is found in the blood plasma, it does not appear in the urine because of

- (A) osmosis
- (B) secretion
- (C) ultrafiltration
- (D) selective reabsorption

31. Which of the following correctly describes movement in plants?

- (A) Irreversible, whole movement
- (B) Reversible, whole movement
- (C) Irreversible, growth movement
- (D) Reversible, growth movement

Item 32 refers to the following diagrams.

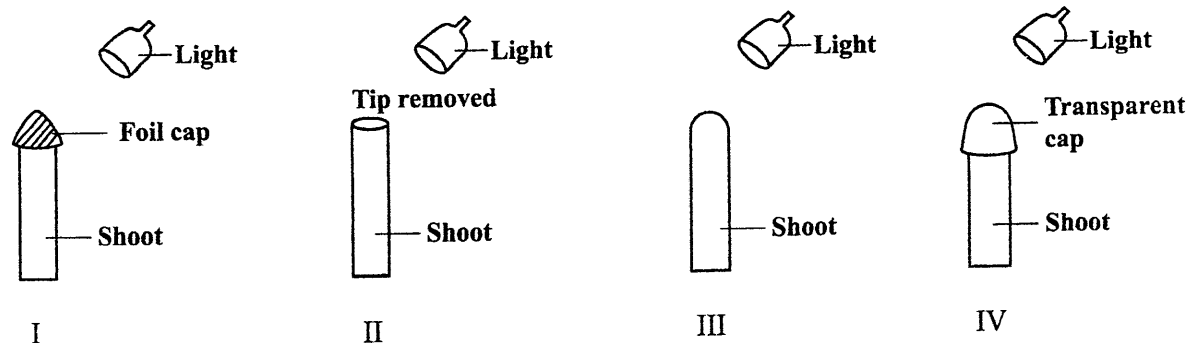


32. Which of the following advantages does the ability to move about give to the chicks compared to the seedlings?

- I. Less competition for space
- II. Access to a variety of foods
- III. Opportunity to use more energy

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

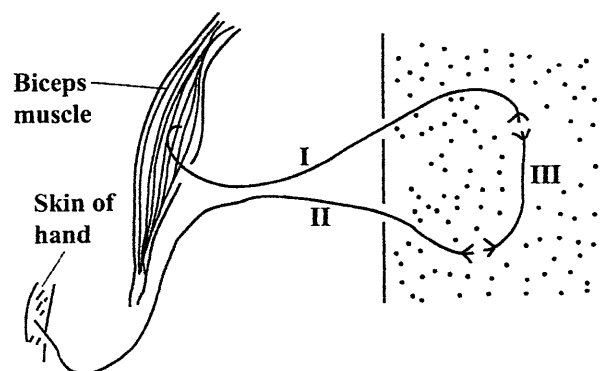
Item 33 refers to the following diagrams illustrating an experiment.



33. After a few days the TALLEST shoot would MOST likely be

- (A) I
- (B) II
- (C) III
- (D) IV

Item 34 refers to the following diagram.



34. The numbered structures are

- | | I | II | III |
|-----|----------|----------|------------------|
| (A) | receptor | effector | relay neurone |
| (B) | receptor | relay | effector neurone |
| (C) | effector | relay | receptor neurone |
| (D) | effector | receptor | relay neurone |

35. The vital centres for the control of heartbeat and respiration are located in the

- (A) cerebral hemispheres
- (B) medulla
- (C) cerebrum
- (D) cerebellum

36. Which of the following is a long-term effect of alcohol abuse?

- (A) Slurring of speech
- (B) Increased urination
- (C) Cirrhosis of the liver
- (D) Dilation of blood vessels

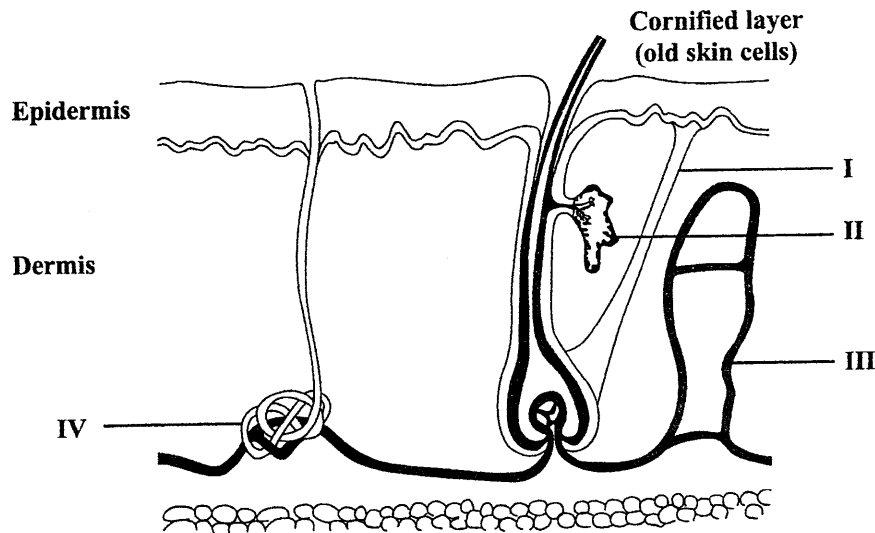
37. Myrie is reading a book. When she looks out of the window to focus on a tree, the lens in her eye becomes

- (A) fatter
- (B) thinner
- (C) more concave
- (D) more convex

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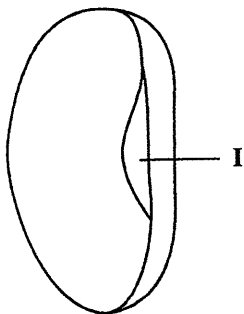
Item 38 refers to the following diagram of a section of human skin.



38. Which structure is NOT involved in temperature regulation?

- (A) I
- (B) II
- (C) III
- (D) IV

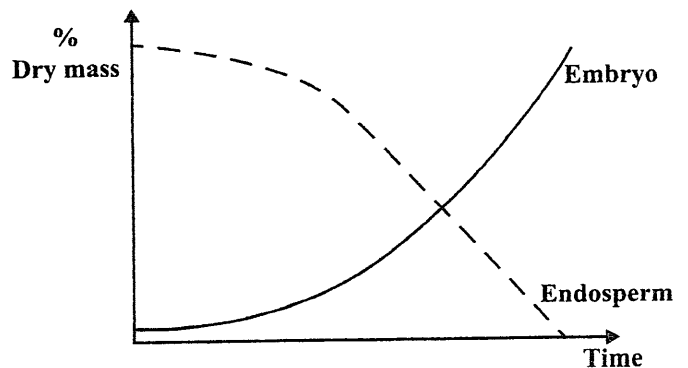
Item 39 refers to the following diagram of a seed.



39. The part labelled I represents the

- (A) hilum
- (B) testa
- (C) micropyle
- (D) position of the radical

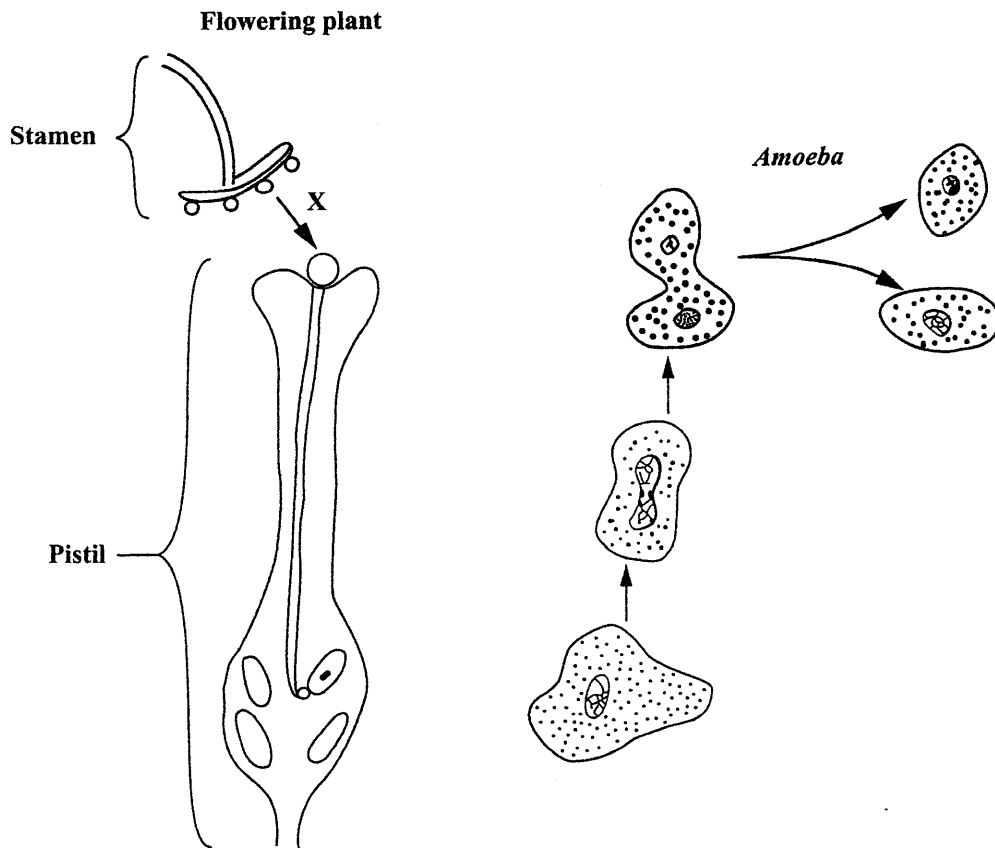
Item 40 refers to the following graph which shows changes in dry mass of embryo and endosperm during germination.



40. From the graph, it is true to say that

- (A) the embryo begins to grow as the endosperm increases in mass
- (B) the embryo and the endosperm decrease in mass at the same rate
- (C) the embryo and the endosperm increase in mass at the same rate
- (D) as the embryo grows, the food reserves are gradually depleted

Items 41–42 refer to the following diagrams which represent events involved in reproduction in a flowering plant and an amoeba.



41. Which of the following statements is true for BOTH methods of reproduction?

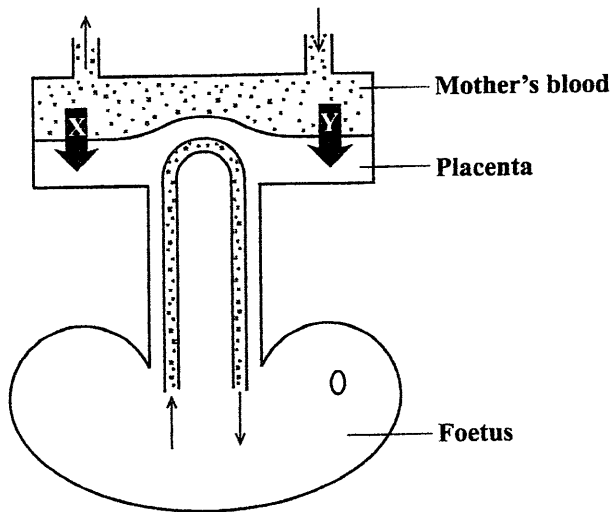
- (A) Cells divide by mitosis.
- (B) Offspring are identical to each other.
- (C) Cells divide by both mitosis and meiosis.
- (D) Both male and female parts are required for reproduction.

42. The event taking place at X is

- (A) growth
- (B) dispersal
- (C) pollination
- (D) fertilization

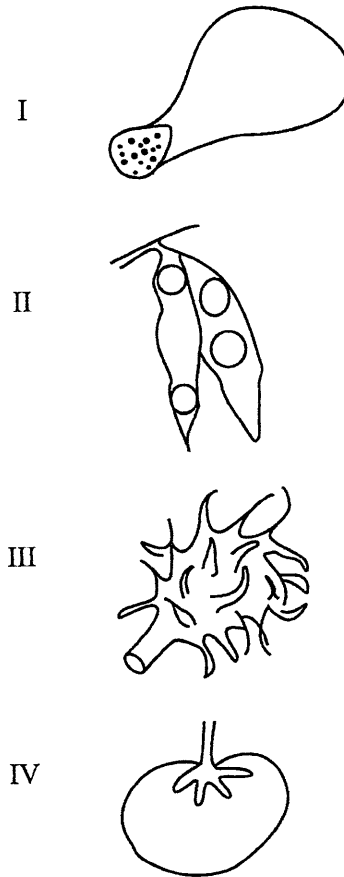
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Item 43 refers to the following diagram which shows the relationship between the blood of a foetus and the blood of a mother.



43. Which substances should NOT move in the direction shown by the arrows labelled X and Y?
- (A) Calcium and glucose
(B) Antibodies and calcium
(C) Amino acids and glucose
(D) Progesterone and fibrinogen
44. A method of contraception that protects against both disease and pregnancy is
- (A) the pill
(B) a condom
(C) spermicide
(D) an intrauterine device
45. Fertilization normally occurs in the
- (A) uterus
(B) vagina
(C) oviduct
(D) ovaries

Item 46 refers to the following diagrams of some fruits/seeds.



46. In which of the following is each fruit or seed matched with its correct method of dispersal?

	I	II	III	IV
(A)	Animal	Wind	Mechanical	Man
(B)	Mechanical	Man	Wind	Animal
(C)	Water	Mechanical	Wind	Man
(D)	Wind	Mechanical	Animal	Man

Items 47–48 refer to the following organisms.

- (A) Host
- (B) Vector
- (C) Parasite
- (D) Pathogen

In answering Items 47–48, match the following descriptions to one of the options above. Each option may be used once, more than once or not at all.

47. An organism which causes a disease
48. An animal which carries a disease-causing organism but is not adversely affected by it.
49. Which of the following methods would be MOST appropriate for the treatment of chronic hypertension?
- I. Diet modification
 - II. Exercise
 - III. Vaccination
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
50. A species consists of members that
- (A) occupy the same habitat
 - (B) reproduce by asexual means
 - (C) interbreed under natural conditions
 - (D) interbreed to produce infertile offspring

51. On a certain Caribbean island where sugar cane is grown, a fungus is accidentally introduced into the island and it destroys the entire crop of sugar cane. Which of the following BEST describes what happened?

- (A) The plants were old and therefore more susceptible to the fungus.
- (B) The plants were not of a hardy enough variety to withstand the fungus.
- (C) The plants were planted too close together so the fungus spread easily.
- (D) There was no genetic variation among the plants so none could survive the attack.

52. If a diploid organism has two different alleles for the same gene, it is described as being

- (A) dominant
- (B) homozygous
- (C) heterozygous
- (D) recessive

53. Which of the following processes will result in genetically identical offspring?

- (A) Cloning
- (B) Mutation
- (C) Fertilization
- (D) Crossing over

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Item 54 refers to the following diagram which represents a cross between two plants, heterozygous for a particular characteristic.

Female \ Male	r	I
R		
r	II	III

54. Which of the following completes the diagram at I, II and III?

- | | I | II | III |
|-----|---|----|-----|
| (A) | r | Rr | rr |
| (B) | R | rr | Rr |
| (C) | r | rr | Rr |
| (D) | R | Rr | rr |

55. Which of the following terms refers to visible characteristics of living things?

- (A) Gene
(B) Genotype
(C) Phenotype
(D) Chromosome

56. Haemophilia is a sex-linked trait caused by a recessive allele. Which of the following are reasons why the disease is expressed in more men than women?

- I. It is caused by a recessive allele carried on the X chromosome.
II. There is only one X chromosome in males.
III. Women have two X chromosomes so that the recessive allele can be masked by the dominant allele.
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

57. Natural selection can be defined as the process by which natural populations

- (A) mutate in order to survive
(B) remain well adapted to their habitat
(C) compete with other populations for survival
(D) gradually bring about changes in their habitat

58. Which of the following is true about the process of artificial selection?

- (A) Tissue culture is used to obtain clones.
(B) Traits which benefit the target organism are selected.
(C) Selected traits may not benefit the target organism.
(D) A single beneficial gene is introduced into the cell of the organism.

59. Which of the following have contributed to the many antibiotic resistant strains of bacteria present today?

- I. It is possible for bacteria to mutate to antibiotic resistant forms.
- II. Exposure to an antibiotic causes some bacteria to mutate.
- III. The mutants reproduce forming a new strain of resistant bacteria.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

60. Which of the following would be involved in the production of insulin by genetic engineering?

- I. Transfer of the human insulin gene to a bacterium
- II. A vector to transfer the insulin gene
- III. Production of insulin by the bacterium

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.